

0107E804-3

Candidate's Seat No: \_\_\_\_\_

**IM.Sc. (CS) Sem.-8 Examination  
Deep Learning**

**Time : 3.00 Hours]**

**July-2025**

**[Max.Marks : 70**

**Instructions:**

- Write both the Sections in the separate answer book.
- Both Sections having equal weightage.
- Draw Diagrams wherever necessary.
- Make Assumptions wherever necessary.

**Section – I**

**Q.1 Define the following with example: (Any – 5) [15]**

1. Convolution.
2. Pooling.
3. Stride.
4. Padding.
5. Sigmoid.
6. Relu.

**Q.2 Answer the following: (Any – 4) [20]**

1. Why are activation functions important in deep learning? What are the different types of activation functions, and for what kinds of tasks are they typically used?
2. What are exploding gradients and vanishing gradients in deep learning, and how do they affect model training? In which scenarios do these issues typically occur?
3. Explain what an Artificial Neural Network (ANN) is and how the backpropagation algorithm works?
4. Describe the structure of a Multilayer Perceptron and explain its typical use cases in machine learning tasks?
5. How do bias and variance relate to the curse of dimensionality in model training, and what impact do they have on model performance in high-dimensional datasets?

**Section - II**

**Q.3 Answer in detail: [10]**

1. Explain the architecture and working of Long Short-Term Memory (LSTM) networks in deep learning. How do LSTMs address the vanishing gradient problem found in traditional RNNs? Discuss their key components and typical applications.

ES04-4

OR

1. Explain the architecture and working principle of Recurrent Neural Networks (RNNs) in deep learning. Discuss how RNNs handle sequential data, their limitations, and how these issues impact model performance.

Q.4 Differentiate the following: (Any – 5)

[25]

1. What is the role of dropout and drop connect in neural network training? How do they contribute to noise robustness?
  2. Describe the concept of adversarial training and its significance in improving model robustness.
  3. Discuss the challenges in optimization when training deep neural networks. How can second-order methods help?
  4. Explain the process of building a deep neural network architecture for image classification using TensorFlow.
  5. What is multi-task learning? How does it relate to regularization and model generalization?
  6. Explain the difference between L1 and L2 regularization. How do they affect model training and sparsity?
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